GEMMA-AETHER

Atmospheric Evaluation & Tensor Forecasting Orbit

Project Type: Multimodal Machine Learning / Spatiotemporal Forecasting / Environmental Intelligence

Primary Model: Gemma 4

Embedding Infrastructure:

- EmbeddingGemma 2 (local)
- Gemini Embedding 2 (cloud)

Research Domain: Atmospheric modeling, localized precipitation forecasting, multimodal retrieval, spatiotemporal representation learning.

Project Status: Proposed architecture / implementation planning.

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1. PROJECT OVERVIEW

GEMMA-AETHER is a multimodal atmospheric intelligence and localized weather forecasting system designed to investigate the combination of learned spatiotemporal representations, dual-model embedding infrastructure, historical atmospheric retrieval, and generative reasoning.

The system will ingest historical and near-real-time atmospheric observations, construct structured representations of atmospheric conditions, retrieve historically analogous weather patterns, and generate localized forecasts through a dedicated numerical prediction architecture.

The project will investigate whether combining independently trained local and cloud-based embedding models can improve historical atmospheric pattern retrieval and downstream forecasting performance.

The system will integrate three primary computational components:

1. Gemma 4 — generative reasoning, interpretation, and structured forecast synthesis.
2. EmbeddingGemma 2 — locally deployed multimodal embedding and retrieval.
3. Gemini Embedding 2 — cloud-based multimodal embedding and independent retrieval.

A dedicated spatiotemporal forecasting backbone will process numerical atmospheric tensors and produce quantitative predictions.

The embedding infrastructure will supplement this forecasting architecture rather than replace its numerical modeling capabilities.

2. PRIMARY RESEARCH OBJECTIVES

The project will investigate five primary questions.

RQ1 — Atmospheric Representation

Can historical atmospheric observations be transformed into useful spatiotemporal representations that preserve relationships between meteorological variables, geographic locations, and temporal evolution?

RQ2 — Dual-Embedding Retrieval

Does combining EmbeddingGemma 2 and Gemini Embedding 2 improve retrieval of historically analogous atmospheric conditions compared with either embedding model independently?

RQ3 — Representation Fusion

Which fusion strategy produces the strongest retrieval performance when combining independently learned embedding spaces?

Candidate approaches:

- Reciprocal Rank Fusion.
- Weighted late fusion.
- Learned projection layers.
- Cross-encoder reranking.
- Supervised representation alignment.

RQ4 — Forecasting Performance

Does retrieval-augmented forecasting improve localized precipitation prediction compared with a forecasting model operating without historical retrieval?

RQ5 — Computational Efficiency

Can a relatively lightweight forecasting architecture produce useful localized predictions with lower computational requirements than larger numerical forecasting systems?

3. SYSTEM ARCHITECTURE

The system will be organized into six computational layers.

Layer 1 — Atmospheric Data Acquisition

Purpose: Collect historical atmospheric observations and meteorological forecasts.

Initial data sources:

ERA5

- Historical atmospheric reanalysis.
- Temperature.
- Atmospheric pressure.
- Humidity.
- Wind components.
- Geopotential.
- Precipitation-related variables.

NOAA HRRR

- High-resolution numerical weather predictions.
- Regional atmospheric conditions.
- Forecast initialization and lead times.

NOAA MRMS

- Multi-Radar Multi-Sensor precipitation observations.
- Radar-derived precipitation estimates.
- High-resolution precipitation analysis.

Local Weather Stations

- Surface temperature.
- Relative humidity.
- Atmospheric pressure.
- Wind speed and direction.
- Observed precipitation.

Initial geographic focus:
Watauga, Tennessee, with the architecture designed to support arbitrary geographic regions.

Data acquisition must preserve:

- Original timestamps.
- Geographic coordinates.
- Spatial resolution.
- Units.
- Source identifiers.
- Observation versus forecast provenance.
- Missing-data indicators.

Historical reanalysis, forecast products, and direct observations must remain distinguishable.

Layer 2 — Atmospheric Tensor Construction

Purpose: Convert meteorological observations into structured numerical inputs.

Represent atmospheric conditions as spatiotemporal tensors.

Proposed input representation:

$X \in R^{(T \times H \times W \times C)}$

Where:

T = Historical temporal sequence length.

H = Spatial grid height.

W = Spatial grid width.

C = Number of meteorological variables.

Example channels:

- 2-meter temperature.
- Surface pressure.
- Relative humidity.
- U-component wind.
- V-component wind.
- Precipitation.
- Geopotential height.

Each sample represents an atmospheric region across a defined historical time window.

Preprocessing requirements:

1. Spatial resampling.
2. Temporal alignment.
3. Unit normalization.
4. Missing-value handling.
5. Geographic masking.
6. Temporal window construction.
7. Train/validation/test partitioning.

Normalization parameters must be calculated using training data only.

Avoid random sample splitting that introduces temporal leakage.

Use chronological evaluation and geographically separated holdouts where appropriate.

Layer 3 — Dual Embedding Infrastructure

This is one of the central experimental components of GEMMA-AETHER.

Two independently trained embedding systems will process corresponding atmospheric representations.

A. EmbeddingGemma 2

Deployment: Local.

Responsibilities:

- Local semantic embedding.
- Multimodal retrieval.
- Historical weather-description indexing.
- Atmospheric image and visualization retrieval.
- Offline retrieval capability.

Initial embedding dimension: 768.

The model supports multimodal representations of text, images, video, and audio.

For the initial implementation, use its text and image capabilities.

Potential inputs:

- Structured meteorological descriptions.
- Atmospheric map imagery.
- Radar imagery.
- Historical weather summaries.

B. Gemini Embedding 2

Deployment: Cloud API.

Responsibilities:

- Independent multimodal embedding.
- Historical atmospheric pattern retrieval.
- Cross-modal similarity search.
- Comparative embedding evaluation.
- Cloud-assisted retrieval.

Initial embedding dimension: 768.

Maintain a separate vector index for this model.

Do not directly compare its vectors against EmbeddingGemma 2 vectors.

The two models have independently learned embedding spaces. Equal dimensionality does not establish geometric alignment.

C. Embedding Metadata

Every embedding record must include:

- Model identifier.
- Model version.
- Embedding dimension.
- Input modality.
- Source dataset.
- Geographic region.
- Timestamp.
- Temporal window.
- Atmospheric variables.
- Preprocessing version.
- Normalization configuration.

This enables reproducible experiments and controlled model comparisons.

Layer 4 — Dual-Embedding Retrieval and Fusion

Purpose: Retrieve historical atmospheric conditions that resemble the current atmospheric state.

Maintain independent retrieval pipelines:

Pipeline A:
EmbeddingGemma 2 → Local Vector Index → Ranked Results.

Pipeline B:
Gemini Embedding 2 → Cloud Vector Index → Ranked Results.

The retrieval systems must first be evaluated independently.

Then implement fusion strategies.

Experiment A: Independent Retrieval

Compare:

- EmbeddingGemma 2 retrieval.
- Gemini Embedding 2 retrieval.

Measure:

- Precision@K.
- Recall@K.
- Mean Reciprocal Rank.
- NDCG@K.
- Retrieval latency.

Experiment B: Reciprocal Rank Fusion

Combine ranked results without assuming vector-space compatibility.

RRF score:

$$RRF(d) = \Sigma\ 1 / (k + rank_i(d))$$

Where:

- d = Retrieved atmospheric record.
- k = Ranking stabilization constant.
- rank_i = Document ranking from an individual retrieval system.

Experiment C: Weighted Late Fusion

Combine independently calculated retrieval scores after calibrating them on a validation set.

Do not combine raw cosine similarities without evaluating their distributions and calibration.

Experiment D: Learned Alignment

Investigate whether a learned projection or alignment network can map corresponding representations into a shared comparison space.

Training pairs will consist of corresponding atmospheric records embedded by both systems.

Evaluate alignment using held-out retrieval benchmarks.

Experiment E: Cross-Encoder Reranking

Use a separate reranking model to assess the relevance of candidate historical atmospheric records.

The goal is to determine whether the combined retrieval pipeline produces more useful historical analogues than either model independently.

Layer 5 — Spatiotemporal Forecasting Backbone

Purpose: Learn atmospheric dynamics and generate quantitative predictions.

This is the numerical prediction engine.

Candidate architectures:

Option A: ConvLSTM

- Spatial feature extraction.
- Temporal sequence modeling.
- Relatively straightforward baseline.

Option B: Spatiotemporal Transformer

- Attention across temporal and spatial representations.
- Flexible long-range dependency modeling.
- Potential for learned atmospheric pattern relationships.

Option C: Video-style Transformer

- Treat atmospheric grid sequences as evolving spatial fields.
- Learn relationships between consecutive atmospheric states.

Option D: Graph Neural Network

- Represent geographic locations as nodes.
- Model relationships between spatially connected locations.
- Investigate irregular sensor networks and geographic dependencies.

Initial recommendation:

Implement ConvLSTM as the baseline, followed by a spatiotemporal Transformer.

Do not introduce multiple complex architectures before establishing baseline performance.

Numerical Prediction Outputs

The forecasting model should produce structured numerical predictions.

Initial targets:

1. Precipitation probability.
2. Expected precipitation accumulation.
3. Precipitation intensity.
4. Near-surface temperature.
5. Relative humidity.
6. Wind speed and direction.

Initial forecast horizons:

- 15 minutes.
- 30 minutes.
- 60 minutes.
- 120 minutes.
- 180 minutes.

Forecast horizons must be adjusted to match the temporal resolution and predictive capabilities of the available training data.

Retrieval-Augmented Forecasting

The forecasting pipeline will optionally receive retrieved historical atmospheric analogues.

Inputs:

- Current atmospheric tensor.
- Recent temporal observations.
- Retrieved historical atmospheric patterns.
- Retrieval relevance scores.
- Geographic metadata.

The forecasting model will learn whether historical analogues improve predictions.

Implement two experimental conditions:

A. Forecasting without retrieval.

B. Forecasting with dual-embedding retrieval.

This comparison is essential for determining whether the embedding infrastructure contributes measurable predictive value.

4. GEMMA 4 INTEGRATION

Gemma 4 will serve as the generative reasoning and interpretation component.

It will not initially be responsible for directly predicting raw atmospheric tensors.

Its responsibilities include:

- Interpreting structured meteorological observations.
- Summarizing retrieved historical analogues.
- Explaining forecast outputs.
- Producing structured forecast reports.
- Identifying inconsistencies between observations and model predictions.
- Supporting interactive queries through the Observatory interface.

Gemma 4 will receive structured numerical outputs from the forecasting subsystem.

Example input:

```
{
 “location”: “Watauga, Tennessee”,
 “forecast_horizon_minutes”: 60,
 “precipitation_probability”: 0.87,
 “expected_precipitation_mm”: 4.2,
 “temperature_celsius”: 21.4,
 “relative_humidity”: 86,
 “retrieval_sources”: [
 “embeddinggemma2”,
 “gemini_embedding_2”
 ]
 }
```

Gemma 4 will convert the structured output into a human-readable forecast.

It must not invent numerical values or alter prediction confidence.

Forecast provenance must remain accessible.

Potential future experiment:

Investigate fine-tuning Gemma 4 to interpret structured atmospheric tensors or serialized meteorological sequences.

This is an experimental extension, not a prerequisite for the initial system.

5. EVALUATION FRAMEWORK

The project must prioritize measurable forecasting performance rather than impressive demonstrations.

Forecasting Metrics

Precipitation:

- Probability of Detection.
- False Alarm Ratio.
- Critical Success Index.
- Brier Score.
- Reliability diagrams.

Continuous variables:

- Mean Absolute Error.
- Root Mean Squared Error.
- Bias.
- Continuous Ranked Probability Score, where probabilistic outputs are available.

Spatial forecasting:

- Fractions Skill Score.
- Spatial displacement error.
- Neighborhood-based precipitation verification.

Retrieval Metrics

- Recall@K.
- Precision@K.
- Mean Reciprocal Rank.
- NDCG@K.
- Retrieval latency.

Computational Metrics

- Model inference latency.
- Embedding generation latency.
- Memory consumption.
- GPU utilization.
- CPU utilization.
- Cloud API cost.
- Throughput.

Required Baselines

Compare against:

1. Persistence forecasting.
2. ConvLSTM without retrieval.
3. Spatiotemporal Transformer without retrieval.
4. EmbeddingGemma 2 retrieval.
5. Gemini Embedding 2 retrieval.
6. Dual-embedding retrieval.
7. Dual-embedding retrieval with forecasting.

Where practical, include established numerical weather prediction products as external comparison baselines.

All comparisons must use consistent forecast initialization times, geographic coverage, lead times, and verification observations.

6. OBSERVATORY INTERFACE

Develop an interactive research interface called:

AETHER OBSERVATORY

Its purpose is to expose model behavior, atmospheric evolution, retrieval results, and forecasting performance.

Primary Visualizations

Atmospheric Vector Field

- U/V wind components.
- Wind direction.
- Wind magnitude.
- Spatial flow visualization.

Temperature Field

- Geographic temperature distribution.
- Temporal temperature changes.

Pressure Field

- Surface pressure gradients.
- Pressure evolution.
- Spatial pressure patterns.

Precipitation Forecast

- Probability heatmaps.
- Predicted precipitation accumulation.
- Forecast trajectories.

Embedding Observatory

- EmbeddingGemma 2 retrieval results.
- Gemini Embedding 2 retrieval results.
- Similarity rankings.
- Fused retrieval rankings.
- PCA visualization.
- UMAP exploratory visualization.

Important: PCA and UMAP visualizations are exploratory projections, not proof that individual embedding dimensions correspond to interpretable atmospheric properties.

Forecast Evaluation

- Prediction versus observation.
- Historical error distributions.
- Model comparison.
- Calibration.
- Lead-time performance.

7. IMPLEMENTATION ROADMAP

PHASE 1 — Data Infrastructure

Objectives:

- Establish ERA5 data ingestion.

- Integrate initial NOAA datasets.
- Define geographic boundaries.
- Build atmospheric preprocessing pipeline.
- Construct temporal windows.
- Establish dataset versioning.
- Implement leakage-resistant chronological splits.

Deliverables:

- Dataset ingestion modules.
- Atmospheric preprocessing pipeline.
- Tensor construction utilities.
- Dataset documentation.
- Reproducible data preparation scripts.

Acceptance criteria:
Atmospheric data can be consistently transformed into validated spatiotemporal tensors.

PHASE 2 — Baseline Forecasting

Objectives:

- Implement persistence baseline.
- Implement ConvLSTM.
- Establish training pipeline.
- Implement validation and testing.
- Establish forecasting metrics.

Deliverables:

- Initial forecasting model.
- Training scripts.
- Evaluation reports.
- Baseline performance measurements.

Acceptance criteria:
The system produces reproducible quantitative forecasts and reports performance against persistence.

PHASE 3 — Dual-Embedding Infrastructure

Objectives:

- Deploy EmbeddingGemma 2 locally.

- Integrate Gemini Embedding 2 API.
- Implement independent embedding pipelines.
- Create separate vector indexes.
- Establish embedding metadata schema.
- Implement historical atmospheric retrieval.

Deliverables:

- Local embedding service.
- Cloud embedding service.
- Independent vector indexes.
- Retrieval API.
- Retrieval evaluation suite.

Acceptance criteria:
Both embedding systems can independently retrieve historical atmospheric records with measurable performance.

PHASE 4 — Embedding Fusion Research

Objectives:

- Implement Reciprocal Rank Fusion.
- Implement weighted late fusion.
- Evaluate ranking calibration.
- Investigate learned alignment.
- Implement optional reranking.
- Compare independent and fused retrieval.

Deliverables:

- Fusion subsystem.
- Retrieval comparison experiments.
- Benchmark reports.
- Reproducible experimental configurations.

Acceptance criteria:
The system can quantitatively establish whether fusion improves retrieval quality.

PHASE 5 — Retrieval-Augmented Forecasting

Objectives:

- Integrate historical retrieval into the forecasting backbone.

- Implement retrieval-conditioned prediction.
- Compare retrieval-assisted and non-retrieval models.
- Evaluate multiple forecast horizons.
- Conduct geographic and temporal holdout experiments.

Deliverables:

- Retrieval-augmented forecasting model.
- Comparative evaluation.
- Ablation studies.
- Forecast performance reports.

Acceptance criteria:
The contribution of historical retrieval to forecasting performance is measurable.

PHASE 6 — Gemma 4 Integration

Objectives:

- Integrate Gemma 4 inference.
- Define structured forecast schemas.
- Implement forecast interpretation.
- Implement historical analogue summaries.
- Implement natural-language queries.
- Preserve numerical forecast provenance.

Deliverables:

- Gemma 4 inference adapter.
- Structured output pipeline.
- Forecast explanation interface.
- Interactive querying system.

Acceptance criteria:
Gemma 4 can interpret and explain numerical predictions without fabricating forecast measurements.

PHASE 7 — AETHER OBSERVATORY

Objectives:

- Develop interactive geographic visualization.
- Implement atmospheric vector fields.
- Visualize prediction trajectories.

- Display retrieval results.
- Display model evaluation.
- Integrate model telemetry.

Deliverables:

- AETHER Observatory dashboard.
- Interactive atmospheric maps.
- Embedding visualization.
- Forecast comparison interface.
- Model performance dashboard.

Acceptance criteria:
Users can inspect atmospheric inputs, model predictions, retrieved analogues, and evaluation results.

PHASE 8 — Optimization and Research Validation

Objectives:

- Optimize inference latency.
- Evaluate local deployment.
- Investigate quantization.
- Evaluate geographic generalization.
- Conduct controlled ablation studies.
- Document limitations and failure cases.

Deliverables:

- Optimization report.
- Final evaluation.
- Reproducible experiments.
- Technical documentation.
- Research findings.

Acceptance criteria:
The system has reproducible results and clearly documented performance boundaries.

8. PROPOSED SOFTWARE STACK

Machine Learning

- Python.

- PyTorch.
- Hugging Face Transformers.
- Hugging Face Datasets.
- Scikit-learn.

Model Infrastructure

- Gemma 4.
- EmbeddingGemma 2.
- Gemini Embedding 2.
- Google GenAI SDK.

Atmospheric Data

- xarray.
- Zarr.
- NetCDF.
- NumPy.
- Pandas.
- Dask.

Retrieval

- FAISS or Qdrant.
- Separate embedding indexes.
- Custom retrieval fusion module.

Visualization

- Gradio or Streamlit for initial development.
- Plotly.
- Matplotlib.
- Cartographic visualization tools.

Experiment Tracking

- MLflow or Weights & Biases.
- Versioned datasets.
- Configuration-driven experiments.
- Reproducible evaluation scripts.

9. PROPOSED PROJECT STRUCTURE

```
gemma-aether/
│
├── README.md
├── pyproject.toml
├── configs/
│   ├── data.yaml
│   ├── embedding.yaml
│   ├── forecasting.yaml
│   └── evaluation.yaml
│
├── data/
│   ├── ingestion/
│   ├── preprocessing/
│   ├── normalization/
│   └── tensor_construction/
│
├── embeddings/
│   ├── embeddinggemma/
│   ├── gemini_embedding/
│   ├── indexing/
│   └── metadata/
│
├── retrieval/
│   ├── local_retrieval.py
│   ├── cloud_retrieval.py
│   ├── fusion.py
│   ├── reranking.py
│   └── evaluation.py
│
├── forecasting/
│   ├── baselines/
│   ├── convlstm/
│   ├── transformers/
│   ├── prediction_heads/
│   └── training/
│
├── gemma/
│   ├── inference.py
│   ├── forecast_interpreter.py
│   └── schemas.py
│
├── evaluation/
│   ├── precipitation/
│   ├── spatial/
```

```
│   ├── retrieval/
│   ├── calibration/
│   └── benchmarks/
│
├── observatory/
│   ├── dashboard/
│   ├── atmospheric_maps/
│   ├── vector_fields/
│   └── embedding_visualization/
│
├── experiments/
│
├── tests/
│
└── docs/
├── architecture.md
├── data_provenance.md
├── model_cards.md
└── research_log.md
```

10. ENGINEERING CONSTRAINTS

The following requirements are mandatory.

1. Never represent generated weather descriptions as actual numerical forecasts.
2. Maintain separate embedding spaces and indexes for independently trained embedding models.
3. Do not assume identical embedding dimensions imply geometric compatibility.
4. Preserve the distinction between historical observations, reanalysis, and model-generated forecasts.
5. Prevent temporal leakage during dataset construction and evaluation.
6. Record model versions and preprocessing configurations.
7. Keep API credentials server-side.
8. Make the cloud embedding dependency optional so local retrieval remains operational.
9. Evaluate forecasting performance against established baselines.
10. Clearly distinguish implemented, experimental, and planned capabilities.
11. Preserve forecast uncertainty and calibration information.
12. Do not claim superiority over operational meteorological systems without controlled empirical evidence.

11. INITIAL DEVELOPMENT PRIORITIES

Do not attempt to implement the entire architecture simultaneously.

The first development milestone should establish three things:

A. Can we reliably ingest and preprocess atmospheric observations?

B. Can we train a baseline model that predicts future atmospheric conditions?

C. Can EmbeddingGemma 2 and Gemini Embedding 2 retrieve historically analogous atmospheric states?

Only after these components work independently should we implement retrieval fusion and retrieval-augmented forecasting.

The initial research milestone is not a polished weather application.

It is a reproducible experimental system capable of establishing whether the proposed architecture provides measurable improvements.

PROJECT VISION

GEMMA-AETHER aims to explore a modular atmospheric intelligence architecture combining numerical spatiotemporal learning, multimodal embedding infrastructure, historical pattern retrieval, and generative reasoning.

Its central contribution is the investigation of dual-embedding retrieval as an additional information pathway for localized atmospheric forecasting.

The long-term objective is a geographically scalable system capable of analyzing atmospheric conditions, retrieving analogous historical patterns, forecasting localized weather events, and exposing its predictions through an interactive research observatory.

GEMMA-AETHER: Atmospheric Evaluation & Tensor Forecasting Orbit.

An experimental platform for learning, retrieving, and forecasting atmospheric dynamics.